

Ahmed Sameh El-Sawalhy Full Stack Software Engineer

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🔗 sawalhy.github.io/portfolio

</> Skills

Languages: C#, JavaScript, TypeScript, Python

Backend: .NET, Node.js, Express.js, NestJS, Zod, SQL, Prisma, Entity Framework, Redis, OAuth2

Frontend: React, Redux, Capacitor, Razor, Tailwind CSS, Bootstrap, Ant Design

Cloud & DevOps: Linux, AWS, GCP, Docker, Jenkins, Git, Airflow, Github Actions

Testing & QA: Jest, Selenium

{ } Professional Experience & Projects

Feb 2024 – Present

GISCON, Software Engineer

- **Collaborated remotely** with the Kuwait-based Product Owner and engineers to gather user requirements, implement features, and resolve **10+ production incidents** with rapid response and resolution.
- **Maintained and enhanced** over 4 critical web applications (.NET MVC, React) for Kuwait's Environment Public Authority (EPA) and the Egyptian Environmental Affairs Agency. Delivered UI/UX improvements, third-party API Integration, new feature deployments, performance optimizations, while writing Clean-code and following coding standards and best practices
- **Led the development** of a Capacitor-based mobile app for hazardous waste management, eliminating field reporting errors and reducing processing time.
- **Integrated OpenAI's ChatGPT** to create an AI-powered conversational assistant for spatial searches and geolocation queries, improving user experience and reducing manual support workload.
- **Proposed and implemented** Airflow health checks and failure alerts (heartbeat, task-fail) via Discord webhooks; eliminated silent DAG failures and cut time-to-detect from hours to minutes across license-expiry and integration pipelines.
- **Built** an air quality analytics **chatbot** on top of a large SQL Server dataset using Node.js/Express and an OpenAI query layer, translating natural-language questions into reliable data requests and returning structured results with clear user-facing summaries.

Jun 2023 – Jan 2024

ACS, Software Engineer

Worked on the backend of a family planning software system in partnership with Egypt's Ministry of Health. The platform supported the distribution and administration of multiple contraceptive types, used by ministry administrators, nurses, and doctors across Egypt.

Oct 2021 – Jan 2022

Valeo - GISACC, Software Engineering Intern

Jun 2022 – Sep 2022

Masarat Misr, Software Engineering Intern

Oct 2022 – Dec 2022

Valeo - GISACC, Software Engineering Intern

☆ Selected Projects

TimeSync – Multi-Timezone Tracker

Full-stack React/TypeScript + Express app that lets remote teams track 400+ timezones with real-time clocks, solar day/night timelines, JWT-secured user dashboards, and Neon/PostgreSQL persistence for a responsive, mobile-friendly experience.

Opinion Pilot

Built an automated system that analyzes **public statements** to infer an **individual's leanings on sensitive topics**. Developed a hybrid **NLP** pipeline using **fine-tuned transformers** and **LLM** reasoning to classify stances, aggregate insights, and generate clear, interpretable profiles for use in media and public-appearance screening.

Traffic Data Scraper

Built a traffic surrogate data pipeline on **different cloud provider** to programmatically collect live travel-time estimates from **public map services**. **Produced a historical dataset and a Tableau dashboard** that provided coverage for **non-instrumented corridors** and informed traffic models and operational decisions.

Price-Intel E-commerce Scraper

Web interface to **track** prices across multiple **products** and multiple **retailers** with Scheduled runs that flags price changes automatically.

🎓 Education

2023

B.Sc. Computer & Communications Engineering, Cairo University